MSC DATA SCIENCE PROJECT BOOK

Abstract, Dataset, Features and Daily Work Record

Student Success Analytics and Early Warning System

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Programme: MSc Data Science

Project status: Easy-language record of completed project work

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Human review is required. The system supports academic intervention; it does not make decisions about students.

1. Project Abstract

This project is called Student Success Analytics and Early Warning System. The main purpose is to help faculty members identify students who may need academic support at an early stage. The system studies information available during the course, such as assessment submission, marks, previous attempts, study credits, and activity in the online learning system.

The project uses machine learning to estimate the chance of a later Fail or Withdrawn result. It places each record in Low, Medium, or High support risk. These levels are only used to arrange faculty review. They are not final decisions and they do not mean that a student will definitely fail. Faculty members must check the record and talk to the student before choosing any support.

Different models were tested, and a calibrated Random Forest model was selected using validation data. On the separate test data, the model obtained ROC-AUC 0.761 and precision-recall AUC 0.711. The High group found 45.5% of the later adverse outcomes, and 72.5% of High-group records later had an adverse outcome. Medium and High together found 74.5% of adverse outcomes. The project also includes SHAP explanations, fairness checks, support suggestions, and dashboards.

2. Where We Got the Data

We used the Open University Learning Analytics Dataset, commonly called OULAD. The original data was published by The Open University and is anonymized. It can be downloaded from the official OULAD source, and copies are also available on Kaggle. If a Kaggle copy is used, the original Open University source must still be mentioned.

The dataset contains seven connected CSV files:

- studentInfo.csv - student, module, previous attempt, credits, and final result information.
- studentRegistration.csv - registration and withdrawal dates.
- assessments.csv - assessment date, type, and weight.
- studentAssessment.csv - student submission date and score.
- studentVle.csv - dated online-learning clicks for each student.
- vle.csv - information about online-learning activities.
- courses.csv - module presentation information.

We joined these files and created one row for one student in one module presentation. We used only the information available up to day 40. The prepared data has 27,042 rows.

3. Output We Are Predicting

The model predicts a later adverse academic outcome. Fail and Withdrawn are placed in the adverse group. Pass and Distinction are placed in the successful group. Final results are used only to train and test the model; they are not used as input features.

Support level	Simple meaning
Low	The record has a lower model-estimated need for review.
Medium	The record may need a normal academic check-in.
High	The record should be reviewed first by faculty.

4. Features We Are Taking

Feature group	Features used in simple words
Previous study	Number of previous attempts and date of registration.
Study load	Number of credits studied.
Assessment work	Assessments due, submitted, late, and completed by day 40.
Assessment marks	Average score, minimum score, weighted score, and submitted assessment weight.
Online activity	Total clicks, active days, clicks in the last 7 and 14 days, and recent active days.
Activity change	Previous-week clicks, change in clicks, and days since the last activity.
Course details	Module code and presentation season.

Gender, age band, disability, highest education, and deprivation band are not used to make the prediction. They are kept separately only to check whether the model shows different error patterns for different groups.

Important note: OULAD does not have normal classroom attendance, GPA, or backlog columns. We did not create false values for these fields. Online clicks are treated as online engagement, not as classroom attendance.

5. Simple Project Method

Step	Activity
1	Understand the student-support problem and decide the project scope.
2	Collect and study the OULAD CSV files.

Step	Activity
3	Clean the data and join the related tables.
4	Create day-40 assessment and online-activity features.
5	Check missing values, duplicates, data types, and distributions.
6	Split the data by time into training, validation, and testing groups.
7	Train and compare four classification models.
8	Select model thresholds using validation data only.
9	Evaluate the final model on untouched test data.
10	Use SHAP to explain model contributions.
11	Check fairness indicators for approved groups.
12	Create faculty, student, and evaluation dashboards.
13	Prepare Power BI data, measures, theme, and report-page plan.
14	Test the code and prepare project documents.

6. Day-to-Day Work Record

The following record is written by work session because exact calendar dates were not recorded in the project files. Dates can be added in the left column of the physical project book.

Work Sessions 1-10

Session	Work completed
Day 1	Selected the topic and wrote the problem statement, aim, and project scope.
Day 2	Studied the given data analytics and preprocessing steps and mapped them to the project.
Day 3	Compared public student datasets and selected OULAD as the main dataset.
Day 4	Created the Python project structure, virtual environment, requirements, folders, and basic documents.
Day 5	Collected the OULAD CSV files and studied their columns and relationships.
Day 6	Joined student, registration, assessment, course, and online-learning tables.
Day 7	Cleaned the data, checked missing values, duplicates, ranges, and invalid records.
Day 8	Fixed day 40 as the prediction point and removed information that was not available by that day.
Day 9	Created assessment features such as due work, submitted work, late work, scores, and completion rates.
Day 10	Created online-activity features such as clicks, active days, recent activity, and activity change.

Work Sessions 11-23

Session	Work completed
Day 11	Performed EDA and prepared data-quality, outcome, module, and cohort summaries.
Day 12	Created the target: Fail or Withdrawn as adverse; Pass or Distinction as successful.
Day 13	Split the data by time into training, validation, and untouched testing cohorts.
Day 14	Trained Dummy Prior, Logistic Regression, Random Forest, and Histogram Gradient Boosting models.
Day 15	Compared the models, calibrated probabilities, and selected Low, Medium, and High thresholds.
Day 16	Evaluated the selected Random Forest on the 2014J test cohort and saved the metrics.
Day 17	Created SHAP global and student-level explanations and checked the main model factors.
Day 18	Performed subgroup fairness checks and documented privacy and ethical rules.
Day 19	Created support indicators, faculty-reviewed intervention options, and the review queue.
Day 20	Built the Streamlit Faculty, Student, and Evaluation dashboard views.
Day 21	Added automated tests and improved run_all.ps1 so the large prepared file can be reused.
Day 22	Prepared the Power BI import workbook, DAX measures, dashboard page design, and colour theme.
Day 23	Prepared the final report, model card, ethics protocol, dataset notes, and current project update.

7. Main Results

Item	Result
Prepared dataset	27,042 day-40 student-module records
Training data	11,628 records
Validation data	6,366 records
Test data	9,048 records
Selected model	Calibrated Random Forest
ROC-AUC	0.761
Precision-recall AUC	0.711
High recall	45.5%
High precision	72.5%
Medium + High recall	74.5%

8. Work Completed and Work Remaining

Completed work:

- Dataset collection, cleaning, joining, preprocessing, and EDA.
- Model training, comparison, calibration, testing, SHAP, and fairness audit.
- Support rules, review queue, Streamlit dashboard, Power BI-ready files, and documentation.
- Thirteen automated tests passed in the latest verified run.

Remaining work:

- Open the prepared model in Power BI Desktop and save the native .pbix dashboard.
- Arrange the report in the university's final dissertation format.
- Prepare final presentation slides and demonstration practice.
- Use local student data only after formal approval and new validation; this is not part of the current prototype.

9. Conclusion

The project follows the full data analytics workflow: problem selection, requirement study, data collection, understanding, cleaning, preprocessing, visualization, model building, evaluation, interpretation, dashboard development, testing, and documentation. The system helps faculty decide which records to review first, but faculty remain responsible for checking the facts and choosing suitable academic support.

Reference

Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open University Learning Analytics Dataset. Scientific Data, 4, 170171. https://doi.org/10.1038/sdata.2017.171